

Notes: Eaton, Green, Roseman, Wu (JFE 2026)

Retail Option Traders and the Implied Volatility Surface

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1 Executive overview

1.1 Research question

The paper asks: **Do retail option traders move option prices in a way that reshapes the implied volatility (IV) surface?** Specifically, does retail demand pressure steepen/flatten: (i) the **term structure** of IV (short vs long maturity), (ii) the **moneyness curve** (OTM vs ITM, i.e. smile/smirk), and (iii) the **call–put IV wedge**?

1.2 Why it is hard (endogeneity)

Retail trading is endogenous to market conditions (volatility, news, sentiment). A simple regression of IV on retail volume risks reverse causality: high IV may attract retail “lottery” demand, and retail trading might itself respond to volatility spikes.

1.3 Identification strategy

The key identification comes from **retail brokerage platform outages** as quasi-exogenous shocks to retail participation. During outages, the option market is open, but many retail investors cannot trade. The paper uses a **difference-in-differences design**:

- *Time dimension*: outage windows vs the same time-of-day windows in the preceding five trading days.
- *Cross-section*: options on stocks with *high predicted retail trading* vs a low-retail control group.

1.4 Headline results

Retail traders are net purchasers of **short-dated, especially out-of-the-money (OTM), options** (and favor calls), while they are net **sellers/writers of longer-dated options**. During brokerage outages, net retail buying pressure falls sharply for short-dated OTM options, and the IV surface moves in a commensurate way: **IV declines on average**, with **larger declines** for the contracts retail usually buys (calls, short maturity, OTM). In contrast, **IV rises for long-dated options** during outages, consistent with reduced retail writing (reduced supply). Thus, retail demand pressure is an economically meaningful determinant of the IV term structure, moneyness curve, and call–put spread.

2 Data and key constructions

2.1 Option trades and retail identification (OPRA)

The main dataset is OPRA option trades (Jan 2019–Jun 2021). Following Bryzgalova–Pavlova–Sikorskaya (2023), retail option activity is proxied by trades flagged in OPRA as:

- **single-leg auction trades**, or
- **automatic executions** with trade size < 5 contracts.

Trades are signed as buys vs sells using the NBBO midpoint rule (Muravyev, 2016): buy if executed above the midpoint, sell if below (midpoint trades are hard to sign).

2.2 Alternative retail proxy (Nasdaq NOTO/PHOTO)

For robustness, the paper also uses Nasdaq option data identifying non-professional “small customer” trades (average trade size < 50 contracts in 10-minute intervals), which can separate opening/closing and purchase/write flows. This is used in robustness panels.

2.3 Implied volatility construction (Black–Scholes inversion)

For each option contract j on underlying stock i at time t , implied volatility $\sigma_{j,t}$ is the value solving:

$$P_{j,t}^{mkt} = BS(S_{i,t}, K_j, r_t, \tau_{j,t}, \sigma_{j,t}),$$

where $P_{j,t}^{mkt}$ is the OPRA NBBO midpoint, $S_{i,t}$ is the TAQ stock NBBO midpoint, r_t is the interpolated zero-coupon risk-free rate, and $\tau_{j,t}$ is time-to-expiration (in years, measured in minutes to market close on expiration date).

To study the IV surface at the stock–time level, the paper volume-weights across the option chain within 15-minute windows:

$$IV_{i,t} \equiv \sum_{j \in \mathcal{J}(i,t)} \omega_{i,j,t} \sigma_{j,t}, \quad \omega_{i,j,t} = \frac{\$Vol_{i,j,t}}{\sum_{\ell \in \mathcal{J}(i,t)} \$Vol_{i,\ell,t}}. \quad (1)$$

Implied volatility is winsorized within outage events at the 99.5th percentile.

2.4 Maturity and moneyness bins

Maturity bins (in calendar days to expiration):

$$\text{Short} : \leq 7, \quad \text{Mid} : 8\text{--}20, \quad \text{Long} : > 20.$$

Moneyness is defined so that positive values correspond to in-the-money for both calls and puts:

$$m_{j,t} \equiv \begin{cases} 1 - \frac{K_j}{S_{i,t}}, & \text{call,} \\ \frac{K_j}{S_{i,t}} - 1, & \text{put.} \end{cases}$$

Moneyness bins:

$$\text{OTM} : m_{j,t} < -0.025, \quad \text{ATM} : -0.025 \leq m_{j,t} \leq 0.025, \quad \text{ITM} : m_{j,t} > 0.025.$$

2.5 Net retail option demand pressure

At the stock–15-minute level, net retail dollar volume is:

$$NetVol_{i,t} \equiv \frac{\$Buy_{i,t}^{ret} - \$Sell_{i,t}^{ret}}{\$Total_{i,t}^{ret}}, \quad (2)$$

i.e. buyer-initiated minus seller-initiated retail dollar volume, scaled by total retail dollar volume.

3 Why retail can reshape the IV surface

A key conceptual point is that option market makers (dealers) typically manage delta risk via the underlying, but face inventory risks in higher-order exposures (gamma/vega/theta), especially when option flows are imbalanced. Thus, **net buying** of options (especially short-dated OTM contracts, which load heavily on gamma/vega for a given premium) can raise dealers' effective risk exposure and require compensation, showing up as **higher implied volatility**. Conversely, if retail traders are net **sellers/writers** in some regions of the surface (e.g., long maturity), they provide supply that can **lower implied volatility** there.

This yields a simple mapping from observed retail preferences (Fig. 2) into testable IV predictions:

- Retail net *demand* (short maturity, OTM, calls) $\Rightarrow$ retail participation *raises* IV there.
- Retail net *supply* (long maturity) $\Rightarrow$ retail participation *lowers* IV there.

Broker outages remove retail participation, so the IV response during outages should be the opposite sign (drops where retail demands, rises where retail supplies).

4 Core empirical models (all regressions in the paper)

4.1 Model (1): outages and option volume / net retail volume

Let t index **15-minute intervals** and d index **days**. For each outage event, the sample includes the outage window on day d and the same time-of-day windows in the previous five trading days. Define:

- $Retail_{i,d-1} = \mathbf{1}\{\text{stock } i \text{ is in top retail-interest quintile over the prior 5 days}\}$,
- $Outage_t = \mathbf{1}\{\text{interval } t \text{ is during the outage window}\}$.

The baseline DID regression is:

$$Trd_{i,t} = \alpha + \beta_1 Retail_{i,d-1} + \beta_2 Outage_t + \beta_3 (Retail_{i,d-1} \times Outage_t) + \gamma_i + \delta_d + \varepsilon_{i,t}. \quad (3)$$

$Trd_{i,t}$ is alternately: (i) log option dollar volume (total or retail, calls or puts), or (ii) net retail demand pressure $NetVol_{i,t}$.

Interpretation of β_3 . β_3 is the DID effect: the change in trading outcomes during outages for high-retail-interest options relative to low-retail-interest options, netting out day effects and time-of-day benchmarking via the control windows.

Standard errors are clustered by stock and day.

4.2 Model (2): outages and implied volatility

Implied volatility is studied using an analogous specification:

$$IV_{i,t} = \alpha + \beta_1 Retail_{i,d-1} + \beta_2 Outage_t + \beta_3 (Retail_{i,d-1} \times Outage_t) + \gamma_i + \delta_d + \varepsilon_{i,t}. \quad (4)$$

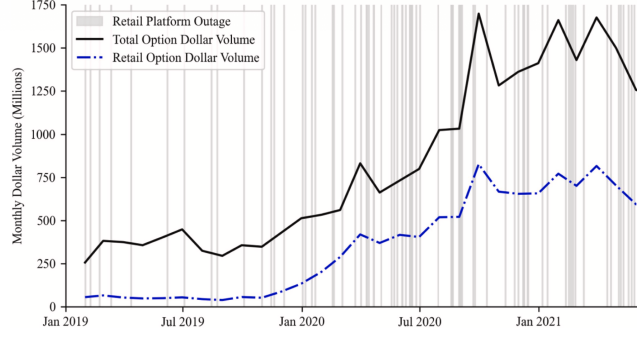


Fig. 1. Monthly Option Trading and Broker Platform Outage Dates. The figure illustrates monthly option dollar volume and brokerage outages. The black solid line plots monthly option dollar volume (calculated as option price $\times$ contract volume aggregated throughout the month), and the blue dashed line plots the measure of retail option volume. Gray bars denote days on which a retail broker experienced an outage during the regular trading hours of 9:30 to 16:00 EST.

$IV_{i,t}$ is the volume-weighted implied volatility across the option chain of stock i within each 15-minute interval.

4.3 Key cross-sectional splits

The paper estimates (3)–(4) on subsamples to map retail preferences to IV-surface movements:

- calls vs puts,
- OTM vs ATM vs ITM,
- short vs mid vs long maturity,
- spread regressions: call–put, OTM–ITM, short–long.

5 Interpreting each table and each figure

5.1 Figure 1: retail option volume growth and outage timing

Fig. 1 shows total option dollar volume and retail option volume (BPS proxy) from Jan 2019 to Jun 2021, with outage dates overlaid. **Interpretation:**

- Retail option trading is a substantial and rising fraction of the overall market in this period.
- Outages are spread throughout the sample and not concentrated in a single month, supporting identification.

5.2 Table 1: summary statistics (what the data look like)

Panel A (stock-day characteristics, retail-dollar-volume weighted). Key takeaways:

- Retail trading is call-heavy: mean call share ≈ 0.76 .
- Contracts traded by retail are short-dated on average (median ≈ 13.6 days).
- IV is high for retail-traded contracts (median ≈ 0.59 ; mean ≈ 0.67).
- Net retail dollar volume is positive on average/median (buying pressure on average), but with wide dispersion.

Table 1
Summary Statistics.

Panel A: Option Characteristics					
	Mean	Standard Deviation	25th Percentile	Median	75th Percentile
Option Dollar Volume	161,846	670,999	5398	12,789	50,996
Retail Dollar Volume	63,778	347,607	952	3939	13,123
Net Retail Dollar Volume	12,915	41,125	-466	1193	4632
Calls	0.76	0.30	0.64	0.89	0.97
Option Price	9.68	17.93	2.15	4.89	10.56
Implied Volatility	0.67	0.47	0.42	0.59	0.81
Moneyness	-0.01	0.06	-0.02	-0.01	0.02
Days to Expiration	19.03	34.65	3.53	13.55	33.61
Panel B: Stock-level Option Trading by 15-minute Periods					
	Mean	Standard Deviation	25th Percentile	Median	75th Percentile
<u>High Retail Interest Stocks</u>					
Implied Volatility	0.68	0.41	0.40	0.57	0.80
Option Trades	5316	6753	1242	3303	6593
Option Dollar Volume	96,897	214,686	10,203	35,509	67,059
Retail Trades	2315	2297	620	1550	3115
Retail Dollar Volume	46,855	103,240	5221	13,499	34,799
Net Retail Dollar Volume	10,346	48,342	288	2195	9493
<u>Low Retail Interest Stocks</u>					
Implied Volatility	0.48	0.34	0.24	0.36	0.61
Option Trades	254	534	36	112	246
Option Dollar Volume	4424	20,420	187	519	2717
Retail Trades	114	271	11	36	108
Retail Dollar Volume	794	1934	58	242	799
Net Retail Dollar Volume	568	3017	-20	0	328

Panel B (15-minute stock-level trading for high vs low retail interest). High retail-interest stocks have far higher option activity and higher IV:

- High-retail: mean IV ≈ 0.68 (median ≈ 0.57) and mean retail trades ≈ 2315 per 15 minutes.
- Low-retail: mean IV ≈ 0.48 (median ≈ 0.36) and mean retail trades ≈ 114 per 15 minutes.

Interpretation: high-retail-interest options are *active* markets, so outage responses are not driven by illiquidity.

5.3 Figure 2: where retail trades on the IV surface

Fig. 2 plots (i) retail dollar volume shares and (ii) net retail volume shares across maturity $\times$ moneyness bins (and separately for calls/puts). **Key message:** retail demand is concentrated in short maturity and OTM/near-the-money, especially calls; retail is more likely to *sell/write* longer-dated options. This figure is the empirical foundation for the paper’s directional predictions about IV changes during outages.

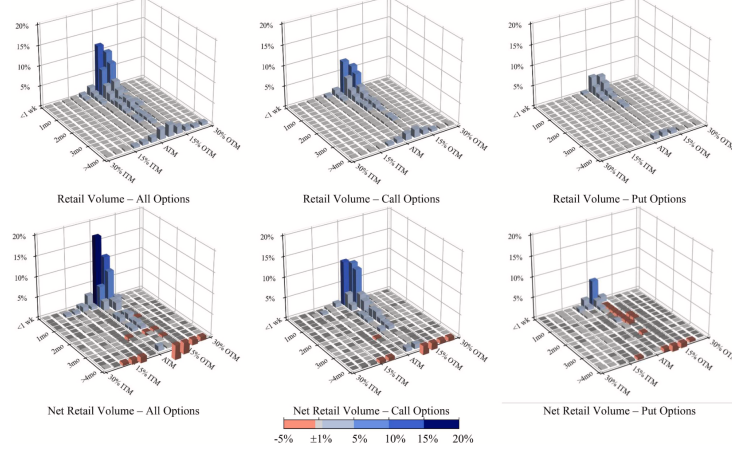


Fig. 2. Retail Option Volume by Expiration and Strike level. The figure plots retail dollar volume share (top row) and net retail volume share (bottom row) for different moneyness and maturity categories. Net retail option volume is defined as the difference between buyer- and seller-initiated dollar volume. Additional plots show volumes for calls and puts separately. The top panels are normalized by total retail dollar volume across all bins. In the bottom panels, the net volume measure for each maturity-moneyness bin is calculated by aggregating the signed net retail dollar volumes and dividing this sum by the total absolute value of the aggregated bins.

Table 2
Retail Broker Outages and Option Dollar Volume.

	Total Option Volume	Retail Option Volume	Retail Call Volume	Retail Put Volume
$\text{Retail}_{i,d-1} \times \text{Outage}_t$	-0.075**	-0.132***	-0.158***	-0.091***
	(-2.378)	(-3.120)	(-3.380)	(-3.142)
$\text{Retail}_{i,d-1}$	0.536***	0.739***	0.717***	0.930***
	(2.732)	(2.899)	(3.332)	(2.903)
Outage_t	0.004	0.059	0.017	0.077
	(0.104)	(1.039)	(0.339)	(0.787)
Observations	632,279	632,279	596,040	273,349
Number of Stocks	842	842	829	671
R-squared	0.124	0.164	0.171	0.243

5.4 Table 2: outages reduce option trading volume (validating the shock)

Table 2 estimates (3) with log dollar volume as $\text{Trd}_{i,t}$. The interaction coefficient β_3 is negative and significant:

- Total option volume: $\beta_3 \approx -0.075$ (significant).
- Retail option volume: $\beta_3 \approx -0.132$ (highly significant).
- Retail call volume drops more than retail put volume (about -0.158 vs -0.091).

Interpretation: outages generate a large contraction in retail option trading precisely where retail is expected to trade, supporting the DID design.

5.5 Table 3: outages change net retail demand pressure in predictable ways

Table 3 estimates (3) with $\text{NetVol}_{i,t}$. **Key pattern:**

- Net retail buying falls for the contracts retail *normally buys*: all options ($\beta_3 \approx -0.083$), calls (≈ -0.103), short maturity (≈ -0.115), OTM (≈ -0.110).
- For long maturity and ITM, the interaction turns positive (e.g. long maturity $\approx +0.048$, ITM $\approx +0.136$), consistent with retail being a net *seller/writer* there in normal times—outages remove that supply.

Table 3
Retail Broker Outages and Net Retail Trader Option Dollar Volume.

	All Options	Calls	Puts	Short Maturity	Mid Maturity	Long Maturity	OTM	ATM	ITM
Retail _{i,d-1} × Outage _t	-0.083*** (-2.702)	-0.103** (-2.561)	-0.041** (-2.070)	-0.115** (-2.274)	-0.122** (-2.189)	0.048** (1.982)	-0.110*** (-2.794)	-0.072** (-2.174)	0.136* (1.804)
Retail _{i,d-1}	0.008 (-0.764)	0.01 (-0.690)	-0.004 (-1.359)	-0.007 (-1.112)	-0.011 (-0.893)	0.004 (1.469)	0.006 (1.162)	0.008 (1.111)	0.008 (0.933)
Outage _t	-0.001 (-0.348)	-0.001 (-0.295)	0.001 (0.374)	-0.001 (-0.284)	-0.002 (-0.519)	-0.001 (-0.086)	-0.001 (-0.237)	0.004 (0.827)	-0.001 (-0.290)
Observations	632,279	596,040	273,349	369,200	348,869	497,534	482,723	471,952	334,180
Number of Stocks	842	829	671	611	714	803	780	745	799
R-squared	0.013	0.013	0.014	0.013	0.014	0.013	0.014	0.013	0.015

Table 4
Retail Broker Outages and Implied Volatility.

	All Options	Calls	Puts	Call - Put
Retail _{i,d-1} × Outage _t	-0.023*** (-3.116)	-0.062*** (-3.016)	-0.019*** (-2.710)	-0.038** (-2.372)
Retail _{i,d-1}	0.104*** (2.201)	0.101*** (2.885)	0.095*** (2.762)	0.102 (0.283)
Outage _t	0.016 (0.953)	0.015 (0.976)	0.026 (1.530)	0.006 (0.392)
Observations	632,279	596,040	273,349	257,008
Number of Stocks	842	829	671	554
R-squared	0.107	0.080	0.085	0.069

Interpretation: outages shift demand pressure across the surface in exactly the direction suggested by Fig. 2.

5.6 Table 4: outages reduce IV (especially for calls) and compress the call-put wedge

Table 4 estimates (4). **Main effect:** for high-retail-interest options, IV falls during outages by about 0.023 on average (economically $\sim 4\%$ relative to the median IV). **Cross-section:**

- Calls: large drop (≈ -0.062).
- Puts: smaller drop (≈ -0.019).
- Call-put differential: negative and significant $\Rightarrow$ call IV falls more than put IV.

Interpretation: retail demand pressure is an important contributor to the call-put IV spread.

5.7 Table 5: outages flatten the moneyness curve (weaker smile/smirk)

Table 5 splits by moneyness bins. **Result:** IV declines strongly for OTM and ATM options, but not for ITM options. The OTM-ITM column confirms a significant differential effect. **Interpretation:** retail demand pressure strengthens the moneyness smile/smirk in normal times; removing retail participation flattens the curve.

Table 5
Retail Broker Outages and the Implied Volatility Moneyness Curve.

	Out of the Money	At the Money	In the Money	OTM-ITM
Retail _{i,d-1} × Outage _t	-0.037*** (-3.436)	-0.020** (-2.393)	0.026 (1.357)	-0.040** (-2.120)
Retail _{i,d-1}	0.100*** (3.380)	0.102*** (2.660)	0.096*** (2.976)	0.005 (0.998)
Outage _t	-0.003 (-0.406)	-0.001 (-0.026)	0.045 (0.664)	-0.039 (-0.215)
Observations	482,723	471,952	334,180	301,253
Number of Stocks	780	745	799	612
R-squared	0.134	0.136	0.160	0.006

Table 6
Retail Broker Outages and Implied Volatility Term Structure.

	Short Maturity	Mid Maturity	Long Maturity	Short - Long
Retail _{i,d-1} × Outage _t	-0.058*** (-2.907)	-0.036** (-2.188)	0.020** (2.384)	-0.063*** (-2.974)
Retail _{i,d-1}	0.139** (2.798)	0.094*** (3.639)	0.054** (2.493)	0.083 (0.888)
Outage _t	0.038 (0.621)	0.009* (1.536)	0.001 (0.113)	0.040 (1.078)
Observations	369,200	348,869	497,534	343,618
Number of Stocks	611	714	803	534
R-squared	0.088	0.070	0.099	0.081

Table 7
Realized Volatility, Implied Volatility, and Brokerage Outages.

	All Outages		Non-Robinhood Outages		Robinhood Outages	
	Stock Volatility	Implied Volatility	Stock Volatility	Implied Volatility	Stock Volatility	Implied Volatility
Retail _{i,d,1} × Outage _i	0.085 (1.387)	-0.023*** (-3.497)	0.099** (2.340)	-0.028** (-2.599)	-0.096** (-2.447)	-0.026** (-2.190)
Retail _{i,d,1}	0.084** (2.255)	0.105*** (2.710)	0.068** (2.313)	0.108** (2.268)	0.113** (2.002)	0.099** (2.309)
Outage _i	0.006 (0.977)	0.016 (0.055)	-0.007 (-1.315)	0.017 (0.704)	0.032 (1.568)	0.012 (0.870)
Observations	632,279	632,279	440,259	440,259	192,020	192,020
Number of Stocks	842	842	707	707	661	661
R-squared	0.063	0.077	0.051	0.068	0.088	0.081

Table 8
Robustness Analysis.

Net Retail Option Volume	Implied Volatility			
	All Options	Call-Put	OTM-ITM	Short – Long
Panel A: Exclude top 30 stocks by option volume				
Retail _{i,d,1} × Outage _i	-0.052** (-2.199)	-0.022*** (-2.677)	-0.038** (-2.195)	-0.054** (-2.484)
Panel B: Exclude morning outages				
Retail _{i,d,1} × Outage _i	-0.028** (-2.379)	-0.037** (-2.320)	-0.032* (-1.827)	-0.031** (-2.071)
Panel C: Identify retail trades only using single-leg auctions				
Retail _{i,d,1} × Outage _i	-0.029* (-1.860)	-0.034*** (-2.725)	-0.047** (-2.321)	-0.035*** (-2.754)
Panel D: Identify retail interest using share of retail dollar volume				
Retail _{i,d,1} × Outage _i	-0.023* (-1.720)	-0.023** (-2.116)	-0.038** (-2.272)	-0.041** (-2.120)
Panel E: Equal-weight across contracts instead of dollar-volume-weight				
Retail _{i,d,1} × Outage _i	-0.032* (-1.720)	-0.028*** (-2.837)	-0.048** (-2.147)	-0.037** (-1.962)
Panel F: Identify retail trading using Nasdaq small customer identifier				
Retail _{i,d,1} × Outage _i	-0.042* (-1.820)	-0.031** (-2.277)	-0.019*** (-3.261)	-0.034* (-1.820)
				-0.020** (-2.305)

5.8 Table 6: outages flatten the term structure (short IV down, long IV up)

Table 6 splits by maturity bins. **Result:**

- Short maturity: large IV drop (≈ -0.058).
- Mid maturity: IV drop (≈ -0.036).
- Long maturity: IV *increases* ($\approx +0.020$).
- Short–long differential is strongly negative.

Interpretation: retail participation tends to *steepen* the IV term structure (push up short IV and push down long IV via writing); outages remove that influence and therefore flatten/tilt the surface.

5.9 Table 7: implied-volatility effects are not mechanically driven by realized stock volatility

Table 7 repeats the analysis for realized volatility of the underlying stock (trade-based) and IV, and splits outages into Robinhood vs non-Robinhood. A key takeaway is:

- IV falls during outages both for Robinhood and non-Robinhood events.
- Realized stock volatility moves differently across broker types (consistent with prior evidence), so IV changes cannot be fully attributed to contemporaneous changes in realized volatility.

Interpretation: the IV response is consistent with option-market demand pressure rather than only a pass-through from underlying volatility.

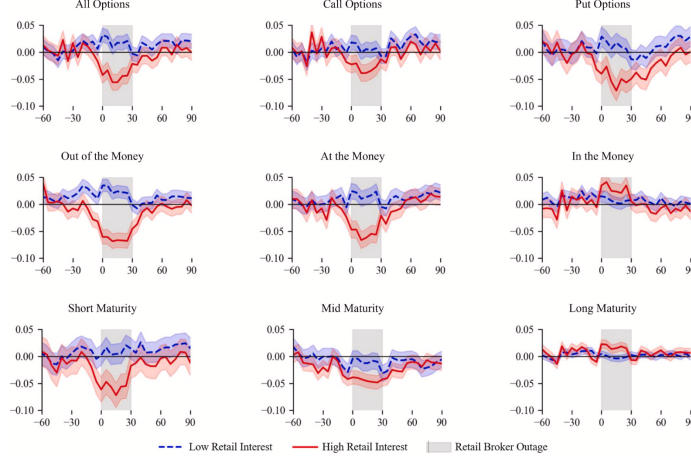


Fig. 3. Implied Volatility around Retail Brokerage Outages. The figure plots abnormal option IV around 30-minute outages. IV is aggregated to the stock level, separately for stocks in the top quintile of retail interest (red solid line) and the remaining quintiles (dotted blue line). For each 5-minute window, abnormal IV is the stock-level IV on the outage day minus IV during the corresponding control period, normalized by the IV standard deviation over the control period. The control period is defined as the same intra-day window over the five trading days preceding the outage. Statistical significance is assessed by regressing abnormal IV on stock and date fixed effects, with 5% confidence intervals in shaded bands.

5.10 Table 8: robustness

Table 8 shows that the main interaction effects remain (direction and significance) under multiple robustness checks:

- excluding top-volume (“meme”) stocks,
- excluding morning outages,
- restricting retail identification to auctions only,
- redefining retail interest as a *share* of total volume,
- equal-weighting rather than dollar-volume weighting,
- using Nasdaq “small customer” data.

Interpretation: the evidence is not driven by one identification choice or a few extreme stocks.

5.11 Table 9: removing the static-quote filter

A potential concern is stale quotes during outages. Table 9 reconstructs the sample without dropping contracts that have no quote changes during the outage window. The main interaction effects remain strongly negative for IV and consistent for cross-sectional differences and net volume.

Interpretation: the results are unlikely to be an artifact of stale quotes.

5.12 Figure 3: event-time abnormal IV (pre-trends and causal timing)

Fig. 3 plots abnormal IV around 30-minute outages (5-minute windows), comparing high-retail-interest vs low-retail-interest options. **Key diagnostic patterns:**

- **No pre-trend:** abnormal IV is near zero before outages for both groups.

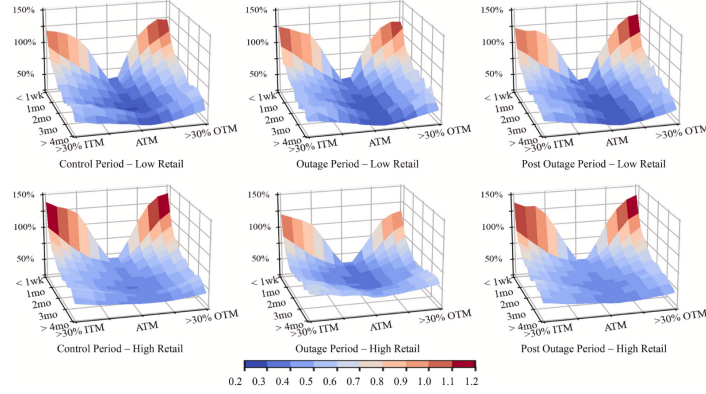


Fig. 4. Implied Volatility Surface around Retail Brokerage Outages. This figure illustrates implied volatility levels in various maturity-moneyness bins before, during, and after brokerage outages for options most favored by retail investors (High Retail) compared to those less favored (Low Retail). Implied volatility for each maturity-moneyness bin is computed by volume-weighting across option chains for each stock day and then averaging across stock days.

- **Sharp discontinuity:** IV moves abruptly during the outage window for high-retail-interest options only.
- **Reversion:** IV returns toward normal after the outage ends.

Interpretation: this is strong graphical evidence supporting a causal effect of retail participation on IV.

5.13 Figure 4: IV surface shifts across maturity $\times$ moneyness

Fig. 4 visualizes the IV surface in bins before/during/after outages for high-retail vs low-retail stocks.

Interpretation: during outages, the surface becomes less “V-shaped” across moneyness and less steep across maturities, with the strongest drops in the short-dated OTM region—exactly where retail net buying is concentrated.

6 Summary

- (1) **Retail demand pressure is a first-order shaper of the IV surface** in modern U.S. equity options, especially in the short-dated OTM region.
- (2) Brokerage outages provide a clean design to separate *participation* effects from endogenous trading.
- (3) The IV term structure response (short down, long up) implies that retail behavior can systematically bias measures of “volatility expectations” inferred from options.
- (4) The call–put and smile/smirk implications matter for work using IV surfaces to back out jump risk, disaster risk, or risk-neutral densities.

7 Conclusion

7.1 What the paper concludes

This paper concludes that **retail option traders are an economically meaningful determinant of the implied-volatility (IV) surface** in U.S. equity options. Using **retail brokerage platform outages** as plausibly exogenous shocks to retail participation, the authors show that when retail traders are suddenly unable to trade, the IV surface shifts in a pattern that closely matches where retail traders typically provide *demand* versus *supply*:

- IV **falls** in the regions of the surface where retail investors are typically **net buyers** (especially **short-dated**, **out-of-the-money**, and **call** options).
- IV **rises** in the region where retail investors are typically **net sellers/writers** (notably **long-dated** options), consistent with outages removing retail *supply*.

Thus, retail option trading pressure does not merely reflect existing volatility; it **causally reshapes** the term structure of IV, the moneyness curve (smile/smirk), and the call-put IV wedge.

7.2 Key quantitative takeaways (from the outage DID estimates)

Let the DID interaction coefficient be the estimate on $\text{HighRetail}_{i,d-1} \times \text{Outage}_t$ (the paper’s main causal object).

- (i) **Outages sharply reduce retail option trading volume (first-stage validation).** Retail option dollar volume drops strongly during outages for high-retail-interest stocks: log-volume falls by about 0.132 (Table 2), and the decline is larger for **calls** (-0.158) than for **puts** (-0.091).
- (ii) **Outages shift net retail demand pressure across the surface in the predicted direction.** Net retail buying pressure falls where retail normally buys: all options (-0.083), calls (-0.103), short maturity (-0.115), and OTM (-0.110) (Table 3). In contrast, net pressure *increases* for long maturity ($+0.048$), consistent with retail being net writers there in normal times (outages remove that retail supply).
- (iii) **Average IV falls during outages, and the drop is concentrated in calls.** For high-retail-interest stocks, IV falls by about 0.023 (Table 4), which the paper interprets as roughly a **4%** decline relative to the median IV of 0.57. The decline is much larger for calls (-0.062) than for puts (-0.019), and the call-put IV wedge compresses (call minus put differential about -0.038).
- (iv) **The moneyness curve flattens because OTM IV falls the most.** IV falls significantly for OTM (-0.037) and ATM (-0.020) options, while ITM IV does not decline in a statistically meaningful way; the OTM-ITM differential is strongly negative (about -0.040) (Table 5).
- (v) **The term structure tilts because short-dated IV falls but long-dated IV rises.** IV falls for short maturity (-0.058) and mid maturity (-0.036), but **increases** for long maturity ($+0.020$); the short-minus-long differential is about -0.063 (Table 6).

7.3 Why retail trading moves the IV surface

The paper’s intuition can be stated as a clean “demand–supply across the surface” mechanism.

1) Retail flow is strongly non-uniform on the IV surface. Retail traders concentrate their **demand** in short-dated options, especially **OTM calls**, while they provide **supply** (writing) in longer-dated contracts. Therefore, retail participation is expected to *raise* IV where retail demands contracts (scarcity from dealers’ perspective) and *lower* IV where retail supplies contracts (extra inventory available to dealers).

2) Dealer intermediation turns flow imbalances into implied-volatility changes. Market makers can hedge delta with the underlying, but option inventory carries risk in higher-order exposures (gamma/vega) and in jump/tail states where hedging is imperfect. When retail is a net buyer of short-dated OTM options, dealers must warehouse more short-dated convexity and vega exposure, and the equilibrium compensation shows up as **higher implied volatility** in those regions.

3) Outages flip the sign of the flow and reveal causality. Broker outages remove retail participation while the market remains open. Thus, during outages:

- where retail is normally net *demand* (short-dated OTM calls), removing retail demand reduces price pressure $\Rightarrow$ IV falls;
- where retail is normally net *supply* (long-dated writing), removing retail supply tightens the market $\Rightarrow$ IV rises.

The observed IV responses by moneyness and maturity match this exact logic, which is why the paper interprets the outage evidence as **causal demand-pressure effects** on the IV surface rather than mere correlation.

4) Why this is not just “underlying realized volatility.” If option IV only moved because stock volatility moved, the implied-volatility patterns should mirror realized-volatility patterns. Instead, the paper finds IV declines during outages across broker types even though realized stock volatility can move differently depending on the broker, suggesting that **option-market demand/supply pressure has its own effect** beyond the underlying market.

7.4 Takeaway

Retail option traders causally reshape the implied-volatility surface: when retail participation is removed by brokerage outages, IV falls sharply where retail normally buys (short-dated, OTM, calls) and rises where retail normally supplies (long-dated options), implying that retail demand pressure is a first-order driver of the IV term structure, moneyness curve, and call–put wedge.